

Dataset Fairness Evaluation Report

Dataset: german.csv
Target Column: credit_risk
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Executive Summary

Executive Summary

Key Fairness Risks

- **Class imbalance severity:**

- **personal_status_sex:** Dominant class *male : single* at **54.8%**; smallest class *male : divorced/separated* at **5.0%** (11× smaller).
- **age:** Dominant class *Young* at **41.1%**; smallest class *Late-Career* at **7.1%** (~6× smaller).

- **Intersectional discrimination (worst affected groups):**

- **male : married/widowed_Early-Middle:** F1 = **0.25**, FPR = **100%**, positive rate = **100%** vs. base rate **33.3%** (extreme over-selection).
- **male : divorced/separated_Mid-Career:** F1 = **0.25**, FNR = **50%**, FPR = **100%** (catastrophic failure, high false positives and false negatives).
- **male : divorced/separated:** FNR = **25%**, FPR = **33.3%**, positive rate **50%** vs. base rate **40%**.
- **Young applicants:** FNR = **15.3%**, FPR = **47.1%**, positive rate **71.0%** vs. base rate **63.4%** (systematic over-selection).

- **Disparity metrics (baseline):**

- ***personal_status_sex*:** Statistical parity difference **0.413**, Disparate impact **0.548** (below 0.8 threshold).
- ***age*:** Statistical parity difference **0.194**, Disparate impact **0.785** (approaching threshold).
- ***Intersectional*:** Statistical parity difference **1.0**, Disparate impact **0.667** (maximal gap).

Mitigation Verdict

- **Reweighting (Balanced + Fair):**

- **Successes:**

- Reduced statistical parity difference for *personal_status_sex* from **0.413** → **0.213** (improved).
- Reduced statistical parity difference for *age* from **0.194** → **0.142** (improved).
- Reduced intersectional statistical parity difference from **1.0** → **0.333** (improved).

- **Failures / Trade-offs:**

- Disparate impact for *personal_status_sex* worsened (**0.548** → **0.767**, still below 0.8).
- Accuracy and F1 declined for several groups (e.g., *male : divorced/separated* accuracy **0.70** → **0.50**, F1 **0.697** → **0.495**).

- FPR increased for multiple groups (e.g., *male : divorced/separated* FPR **33.3% → 66.7%**).

- **Overall: Partial mitigation** — disparities reduced but not resolved, with notable performance degradation for some subgroups.

- **SMOTE (Oversampling):**

- **Successes:**

- Achieved perfect class balance (50/50) and increased dataset size by **+40%**.

- Overall accuracy improved (**0.76 → 0.854**) and overall F1 improved (**0.678 → 0.854**).

- Some subgroup F1 scores improved markedly (e.g., *Early-Middle* age group F1 **0.590 → 0.898**).

- **Failures / Trade-offs:**

- Disparate impact for *personal_status_sex* **worsened (0.548 → 0.202)**, far below 0.8.

- Intersectional disparate impact **worsened (0.667 → 0.162)**.

- Several small groups (e.g., *male : divorced/separated_Early-Middle*) saw F1 collapse (**0.667 → 0.455**) and high FNR (up to **100%**).

- Risk of overfitting due to duplication/synthetic samples and inflated dataset size.

- **Overall: Mixed** — strong overall performance gains but fairness gaps widened for key sensitive attributes; introduced new risks for small intersectional groups.

Bottom Line

- **Reweighting** delivered **moderate, targeted fairness improvements** with fewer extreme side effects but did not fully close disparity gaps.

- **SMOTE** delivered **strong overall performance gains** but **exacerbated fairness risks** for sensitive attributes and small intersectional groups.

- **Recommendation:** Prefer **reweighting or hybrid approaches** (reweighting + calibrated thresholds) over pure oversampling for fairness-critical deployments. Follow up with **intersectional fairness audits, group-calibrated thresholds**, and **adversarial or constrained debiasing** to close remaining gaps without sacrificing subgroup reliability.

Stage 0: Dataset Loading

Tool Used: ``load_dataset``

Analysis

Dataset Summary

The German Credit dataset (``german.csv``) contains **1,000 records** with **21 columns** related to credit risk assessment. Here's a brief overview:

Key Characteristics:

- **Target Variable:** ``credit_risk`` (binary: 0 = good, 1 = bad credit risk)

- **Sample Size:** 1,000 loan applicants

- **Feature Types:** Mix of categorical and numerical attributes

Main Feature Categories:

1. **Financial Status:** `status` (account balance), `amount` (credit amount), `savings` (savings account)
2. **Employment:** `employment\_duration`, `job` (employment type), `personal\_status\_sex`
3. **Credit History:** `credit\_history`, `number\_credits`, `other\_debtors`
4. **Demographics:** `age`, `foreign\_worker`, `personal\_status\_sex`
5. **Loan Details:** `duration` (loan duration), `purpose`, `installment\_rate`
6. **Housing:** `housing` (housing situation), `property` (property ownership)

Sample Observations:

- Credit amounts range from ~1,000 to ~6,000 DM in the preview
- Loan durations vary from 6 to 48 months
- Mix of employment types (skilled employee, unskilled, etc.)
- Both domestic and foreign workers represented
- Various purposes: domestic appliances, retraining, etc.

This dataset is commonly used for studying credit risk prediction and fairness analysis, particularly regarding sensitive attributes like age, gender, and foreign worker status.

Stage 1: Objective Inspection

Objective: Evaluate the dataset 'german.csv' for data quality and fairness issues. Target: credit_risk. Provide a detailed report highlighting any problems found and suggestions for improvement.

Validation: Dataset format compatible (CSV)

Stage 2: Data Quality Analysis

Tool Used: `check\_missing\_data`

Analysis

Summary

The missing data report for german.csv reveals significant data quality issues across 3 columns out of 1000 total rows. The overall missing percentage is 8.20%, but this masks severe problems in specific

columns where missing values exceed 80%. Two categorical columns show extremely high missing rates (90.7% and 81.4%), while one numeric column contains a suspicious value that may represent missing or placeholder data.

Key Findings

- Critical Missing Data in Categorical Columns:** Two columns (`other_debtors` and `other_installment_plans`) have missing rates above 80%, making them nearly unusable for analysis without intervention.
- Suspicious Numeric Value:** The `amount` column contains one instance of value 999 (0.1%), which may represent a placeholder for missing or invalid data.
- Consistent NA Encoding:** Both high-missing columns use "none" as the NA value, suggesting systematic encoding of missing data rather than true null values.

Detailed Analysis

Column-Specific Issues

- amount** (int64)
 - Missing values: 0 (0.0%)
 - Suspicious pattern: Value 999 appears 1 time (0.1%)
 - Issue: Potential placeholder value that should be treated as missing
- other_debtors** (object)
 - Missing values: 907 (90.7%)
 - NA values detected: ["none"]
 - Issue: Extremely high missing rate renders this column nearly useless for analysis
- other_installment_plans** (object)
 - Missing values: 814 (81.4%)
 - NA values detected: ["none"]
 - Issue: Very high missing rate severely limits analytical value

Recommendations

- Immediate Actions:**
 - Investigate the meaning of value 999 in the `amount` column and determine if it should be treated as missing
 - Consider dropping the `other_debtors` column entirely due to 90.7% missing data
 - Evaluate whether `other_installment_plans` can be salvaged or should also be dropped
- Data Cleaning:**
 - Standardize missing value representation across all columns
 - Replace "none" values with proper null/NaN values
 - Consider imputation strategies for the `amount` column if 999 is confirmed as missing

3. Analytical Considerations:

- Any analysis using these columns should account for the extreme missing data
- Results may be biased if missingness is not random
- Consider collecting additional data to supplement these incomplete features

Stage 3: Sensitive Attribute Detection

Tool Used: ``detect_sensitive_attributes``

Analysis

Detected Sensitive Attributes

Column	Reason
personal_status_sex	Combines sex/ gender and marital status, both protected demographic attributes
age	Age is a protected demographic attribute

Column	Description	Sample Values
personal_status_sex	Combines sex/gender and marital status, both pr...	male : single, female : divorced...
age	Age is a protected demographic attribute	67, 22, 49, 45, 53

Stage 3.5: Sensitive Attribute Discretization

Method: auto **Columns Discretized:** 1

age

- **Binning Method:** auto
- **Bin Edges:** [19.0, 30.0, 40.0, 55.0, 75.0]
- **Labels:** Young, Early-Middle, Mid-Career, Late-Career

Bin Distribution::

Bin	Count
Young	411
Early-Middle	315
Mid-Career	203

Late-Career	71
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Agent Reasoning

age

The agent analysed the distribution and semantics of `age` and chose 4 bins: Young, Early-Middle, Mid-Career, Late-Career. Bin edges: [19.0, 30.0, 40.0, 55.0, 75.0].

Stage 4: Imbalance Analysis

Tool Used: `check\_class\_imbalance`

Analysis

Class Imbalance Details

Column	Dominant Value	Percentage	Top Distribution
personal_status_sex	male : single	54.8%	male : single: 54.8%, female : divorced/ separated/ married: 31.0%, male : married/ widowed: 9.2%, male : divorced/ separated: 5.0%
age	Young	41.1%	Young: 41.1%, Early-Middle: 31.5%, Mid-Career: 20.3%, Late-Career: 7.1%

Base Fairness ML Model

- **Algorithm:** Random Forest
- **Test Size:** 0.25
- **Accuracy:** 0.7600
- **Parameters:** `n\_estimators=100`, `max\_depth=None`

Evaluated Fairness Metrics

Sensitive Attribute	Stat Parity Diff	Disparate Impact	Highest Rate Group	Lowest Rate Group
personal + status + sex	0.4130	0.5476	male : married/ widowed	male : divorced/ separated
age	0.1939	0.7854	Early-Middle	Young
personal + status + sex + age	1.0000	0.6667	male : married/ widowed_ Early-Middle	male : divorced/ separated_ Young

Class Imbalance Severity in Sensitive Attributes

personal_status_sex

- Dominant class: male : single at 54.8%
- Distribution skew: 54.8% (male : single), 31.0% (female : divorced/separated/married), 9.2% (male : married/widowed), 5.0% (male : divorced/separated)
- Imbalance severity: High. One category exceeds half the dataset while the smallest category (male : divorced/separated) is 1/11th the size of the dominant category.

age

- Dominant class: Young at 41.1%
- Distribution skew: 41.1% (Young), 31.5% (Early-Middle), 20.3% (Mid-Career), 7.1% (Late-Career)
- Imbalance severity: Moderate to High. The dominant class is nearly 6 times larger than the smallest class (Late-Career).

Fairness Risks and Underrepresented Groups

personal_status_sex

- Underrepresented groups: male : married/widowed (23 samples, 9.2%) and male : divorced/separated (10 samples, 5.0%)
- Risk profile: Small sample sizes amplify variance in performance metrics and increase susceptibility to overfitting on noise.

age

- Underrepresented group: Late-Career (17 samples, 7.1%)
- Risk profile: Limited representation reduces statistical power for reliable fairness assessment in this group.

Impact on Model Bias

Base Rate vs Selection Rate Disparities

personal_status_sex

1. male : single: Base Rate 74.29%, Selection Rate 82.14% (model amplifies positive selection by 7.85 points)
2. female : divorced/separated/married: Base Rate 67.53%, Selection Rate 77.92% (model amplifies positive selection by 10.39 points)

3. male : married/widowed: Base Rate 65.22%, Selection Rate 91.30% (model amplifies positive selection by 26.08 points)
4. male : divorced/separated: Base Rate 40.00%, Selection Rate 50.00% (model amplifies positive selection by 10.00 points)

age

1. Young: Base Rate 63.44%, Selection Rate 70.97% (model amplifies positive selection by 7.53 points)
2. Early-Middle: Base Rate 78.31%, Selection Rate 90.36% (model amplifies positive selection by 12.05 points)
3. Mid-Career: Base Rate 66.67%, Selection Rate 78.95% (model amplifies positive selection by 12.28 points)
4. Late-Career: Base Rate 76.47%, Selection Rate 88.24% (model amplifies positive selection by 11.77 points)

False Negative Rate (FNR) Analysis

personal_status_sex

- FNR values: male : single 9.62%, female : divorced/separated/married 9.62%, male : married/widowed 6.67%, male : divorced/separated 25.00%
- FNR Ratio (Max/Min): $25.00\% / 6.67\% = 3.75$
- Critical observation: male : divorced/separated exhibits the highest FNR at 25.00%, indicating the model fails to select qualified candidates from this group at 3.75 times the rate of the best-performing group.

age

- FNR values: Young 15.25%, Early-Middle 6.15%, Mid-Career 10.53%, Late-Career 0.00%
- FNR Ratio (Max/Min): $15.25\% / 0.00\% = \text{undefined}$ (division by zero)
- Critical observation: Late-Career shows 0% FNR while Young shows 15.25%, indicating extreme disparity in false negative risk. Young applicants face substantially higher rates of being overlooked despite qualification.

Statistical Parity and Disparate Impact

personal_status_sex

- Statistical Parity Difference: 0.413
- Disparate Impact: 0.5476
- Maximum positive rate group: male : married/widowed (91.30%)
- Minimum positive rate group: male : divorced/separated (50.00%)
- Interpretation: The model creates a 41.3 percentage-point gap in positive prediction rates between the most and least favored groups. Disparate impact below 0.8 indicates potential adverse effect.

age

- Statistical Parity Difference: 0.1939
- Disparate Impact: 0.7854
- Maximum positive rate group: Early-Middle (90.36%)

- Minimum positive rate group: Young (70.97%)
- Interpretation: A 19.39 percentage-point gap exists between age groups. Disparate impact of 0.7854 approaches but remains below the 0.8 threshold for concern.

False Positive Rate (FPR) Disparities

personal_status_sex

- FPR range: 41.67% (male : single) to 87.50% (male : married/widowed)
- Critical observation: male : married/widowed experiences FPR of 87.50%, meaning 7 out of 8 negative instances are incorrectly classified as positive, indicating severe over-selection of unqualified candidates from this group.

age

- FPR range: 47.06% (Young) to 77.78% (Early-Middle)
- Critical observation: Early-Middle group shows FPR of 77.78%, indicating the model incorrectly classifies 7 out of 9 negative instances as positive, suggesting systematic over-selection of unqualified candidates in this age bracket.

Bias Amplification Assessment

The model amplifies existing bias across all sensitive attribute groups, as Selection Rate exceeds Base Rate in every category. The amplification is most severe for male : married/widowed (+26.08 percentage points) and Early-Middle age group (+12.05 percentage points). The disparity between base rates and selection rates indicates the model does not merely reflect existing demographic patterns but actively exacerbates positive prediction gaps, particularly for underrepresented groups where small sample sizes may contribute to unstable calibration.

Stage 4.5: Target Fairness Analysis

Tool Used: `analyze\_target\_fairness`

Analysis

Target Variable Rates by Sensitive Group

Sensitive Feature	Group Level	Total Count	Target Distribution
personal_status_sex	male : single	13	1: 53.9%, 0: 46.1%
personal_status_sex	male : married/ widowed	4	1: 75.0%, 0: 25.0%
personal_status_sex	female : divorced/ separated/ married	4	1: 75.0%, 0: 25.0%
age	Early-Middle	7	1: 71.4%, 0: 28.6%
age	Young	11	1: 54.5%, 0: 45.5%
age	Late-Career	1	1: 100.0%

age	Mid-Career	2	1: 50.0%, 0: 50.0%
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Per-Attribute Fairness ML Model

- **Algorithm:** Random Forest
- **Test Size:** 0.25
- **Accuracy:** 0.7600
- **Parameters:** `n\_estimators=100`, `max\_depth=None`

Evaluated Fairness Metrics

Sensitive Attribute	Stat Parity Diff	Disparate Impact	Highest Rate Group	Lowest Rate Group
personal + status + sex	0.4130	0.5476	male : married/ widowed	male : divorced/ separated
age	0.1939	0.7854	Early-Middle	Young

Intersectional Fairness ML Model

- **Algorithm:** Random Forest
- **Test Size:** 0.25
- **Accuracy:** 0.7720
- **Parameters:** `n\_estimators=100`, `max\_depth=None`

Evaluated Fairness Metrics

Sensitive Attribute	Stat Parity Diff	Disparate Impact	Highest Rate Group	Lowest Rate Group
personal + status + sex + age	1.0000	0.6667	male : married/ widowed_ Early-Middle	male : divorced/ separated_ Young

Target Fairness Analysis for Credit Risk

1. Target Distribution Across Demographic Groups

The overall target distribution shows 61.9% positive credit risk (1) and 38.1% negative credit risk (0) across 21 samples. However, distribution varies significantly across sensitive attributes:

By *personal_status_sex*:

- male : single: 53.85% positive rate (7/13) - below overall average
- male : married/widowed: 75.0% positive rate (3/4) - above overall average
- female : divorced/separated/married: 75.0% positive rate (3/4) - above overall average

By age::

- Young: 54.55% positive rate (6/11) - below overall average
- Early-Middle: 71.43% positive rate (5/7) - above overall average
- Mid-Career: 50.0% positive rate (1/2) - below overall average
- Late-Career: 100.0% positive rate (1/1) - significantly above average

2. Disparate Impact Analysis

Single-Attribute Disparities::

1. personal_status_sex shows severe disparate impact:
 - Statistical Parity Difference: 0.413
 - Disparate Impact: 0.5476 (below 0.8 threshold)
 - Positive rate ranges from 50.0% (male : divorced/separated) to 91.3% (male : married/widowed)
 - Gap of 41.3 percentage points between highest and lowest positive rates
2. age demonstrates moderate disparities:
 - Statistical Parity Difference: 0.1939
 - Disparate Impact: 0.7854 (approaching threshold)
 - Positive rate ranges from 52.38% (Young) to 90.36% (Early-Middle)
 - Gap of 37.98 percentage points between groups

Intersectional Disparities::

- Combined attributes show extreme disparity: Statistical Parity Difference of 1.0
- Disparate Impact: 0.6667
- Positive rates span from 0.0% (male : divorced/separated_Young) to 100.0% (male : married/widowed_Early-Middle, male : married/widowed_Young, female : divorced/separated/married_Late-Career)
- Maximum possible gap of 100 percentage points between intersectional groups

3. Intersectional Fairness Analysis

F1 Score Performance by Intersectional Groups::

1. male : divorced/separated_Young: F1 Score 1.0 (perfect performance, n=1)
2. female : divorced/separated/married_Early-Middle: F1 Score 0.88 (n=15)
3. male : married/widowed_Young: F1 Score 0.7143 (n=14)
4. male : single_Mid-Career: F1 Score 0.7259 (n=37)
5. male : single_Late-Career: F1 Score 0.7778 (n=6)
6. male : single_Young: F1 Score 0.6955 (n=38)
7. female : divorced/separated/married_Young: F1 Score 0.6703 (n=40)
8. male : divorced/separated_Early-Middle: F1 Score 0.6667 (n=6)

9. male : single_Early-Middle: F1 Score 0.611 (n=59) - **LOWEST PERFORMANCE AMONG SUBSTANTIAL GROUPS**

10. female : divorced/separated/married_Mid-Career: F1 Score 0.5417 (n=11)

11. male : divorced/separated_Mid-Career: F1 Score 0.25 (n=3)

12. male : married/widowed_Mid-Career: F1 Score 0.4 (n=6)

13. male : married/widowed_Early-Middle: F1 Score 0.25 (n=3) - **LOWEST PERFORMANCE OVERALL**

Critical Finding: male : married/widowed_Early-Middle and male : divorced/separated_Mid-Career both exhibit F1 Scores of 0.25, representing catastrophic model failure for these intersectional groups.

4. Statistical Parity Violations

Base Rate vs Selection Rate Disparities::

1. male : single_Young: Base rate 68.42%, Selection rate 68.42% - parity maintained
2. male : single_Early-Middle: Base rate 84.75%, Selection rate 93.22% - over-selection by 8.47 points
3. female : divorced/separated/married_Young: Base rate 60.0%, Selection rate 70.0% - over-selection by 10.0 points
4. male : married/widowed_Young: Base rate 64.29%, Selection rate 85.71% - over-selection by 21.42 points
5. female : divorced/separated/married_Early-Middle: Base rate 80.0%, Selection rate 86.67% - over-selection by 6.67 points
6. male : divorced/separated_Young: Base rate 0.0%, Selection rate 0.0% - parity maintained
7. female : divorced/separated/married_Late-Career: Base rate 81.82%, Selection rate 90.91% - over-selection by 9.09 points
8. male : single_Mid-Career: Base rate 64.86%, Selection rate 81.08% - over-selection by 16.22 points
9. female : divorced/separated/married_Mid-Career: Base rate 63.64%, Selection rate 81.82% - over-selection by 18.18 points
10. male : married/widowed_Early-Middle: Base rate 33.33%, Selection rate 100.0% - extreme over-selection by 66.67 points
11. male : married/widowed_Mid-Career: Base rate 83.33%, Selection rate 83.33% - parity maintained
12. male : divorced/separated_Early-Middle: Base rate 33.33%, Selection rate 66.67% - over-selection by 33.34 points
13. male : divorced/separated_Mid-Career: Base rate 66.67%, Selection rate 66.67% - parity maintained
14. male : single_Late-Career: Base rate 66.67%, Selection rate 83.33% - over-selection by 16.66 points

Systematic Pattern: 10 out of 14 intersectional groups show over-selection, with male : married/widowed_Early-Middle experiencing the most extreme violation (66.67 percentage point gap).

5. False Negative Rate Disparities

FNR Analysis Across Groups::

1. male : married/widowed_Early-Middle: FNR 0.0% - no rejections
2. female : divorced/separated/married_Early-Middle: FNR 0.0% - no rejections
3. female : divorced/separated/married_Late-Career: FNR 0.0% - no rejections
4. male : married/widowed_Young: FNR 0.0% - no rejections
5. male : single_Late-Career: FNR 0.0% - no rejections
6. male : divorced/separated_Early-Middle: FNR 0.0% - no rejections
7. male : single_Mid-Career: FNR 4.17% - minimal rejections
8. male : single_Early-Middle: FNR 4.0% - minimal rejections
9. male : married/widowed_Mid-Career: FNR 20.0% - moderate rejections
10. female : divorced/separated/married_Young: FNR 16.67% - moderate rejections
11. male : single_Young: FNR 19.23% - moderate rejections
12. female : divorced/separated/married_Mid-Career: FNR 14.29% - moderate rejections
13. male : divorced/separated_Mid-Career: FNR 50.0% - **HIGH REJECTION RATE**
14. male : divorced/separated_Young: FNR 19.23% - moderate rejections

Critical Finding: male : divorced/separated_Mid-Career exhibits 50% FNR, indicating systematic rejection of half the positive cases in this group. Combined with their low F1 Score (0.25), this group faces severe discrimination in credit risk assessment.

6. Risk of Discrimination and Bias

Quantified Bias Indicators::

1. **Intersectional Amplification:** Combined sensitive attributes produce worse disparities than individual attributes alone. Statistical parity difference increases from 0.413 (personal_status_sex alone) to 1.0 (intersectional).
2. **Performance Stratification:** F1 Scores range from 0.25 to 1.0 across intersectional groups, indicating the model works effectively for some demographics while catastrophically failing for others.
3. **Small Sample Vulnerability:** Groups with n=1 (male : divorced/separated_Young) achieve perfect scores, while groups with n=3 (male : married/widowed_Early-Middle, male : divorced/separated_Mid-Career) show worst performance, suggesting instability in minority intersectional categories.
4. **Systematic Over-Selection:** 71.4% of intersectional groups experience positive rate inflation above their base rates, with the most severe case (male : married/widowed_Early-Middle) showing 100% selection rate against 33.33% base rate.
5. **Rejection Pattern Disparities:** FNR ranges from 0% to 50%, with male : divorced/separated_Mid-Career facing rejection rates 2.5x higher than the next worst group and 10x higher than the best-performing groups.

Discrimination Risk Assessment: The model exhibits compound discrimination effects where gender, marital status, and age interact to produce dramatically different outcomes. The combination of low F1 scores, high FNR disparities, and extreme statistical parity violations indicates systematic bias against specific intersectional groups, particularly males who are divorced/separated in mid-career and males who are married/widowed in early-middle age categories.

Stage 5: Recommendations

Recommendations

Top 3 Critical Issues

1. Severe Class Imbalance in Target Variable

The `credit_risk` target shows significant imbalance with 70.0% positive cases (bad credit risk) versus 30.0% negative cases (good credit risk). This 2.33:1 ratio creates bias toward predicting positive outcomes and reduces model sensitivity to the minority class, particularly affecting precision and recall balance for good credit risk predictions.

2. Extreme Missing Data in Critical Columns

Two categorical columns exhibit catastrophic missing data rates: `other_debtors` (90.7% missing) and `other_installment_plans` (81.4% missing). These columns use "none" as a systematic NA encoding rather than true null values, rendering them nearly unusable for analysis without substantial intervention and potentially introducing bias if missingness correlates with credit risk outcomes.

3. Intersectional Fairness Violations in Sensitive Attributes

The combination of `personal_status_sex` and `age` creates severe disparate impact across demographic groups. Statistical parity difference reaches 1.0 (maximum possible) for intersectional groups, with positive prediction rates ranging from 0.0% to 100.0%. Key violations include `male : married/widowed_Early-Middle` (100% positive rate vs 33.33% base rate) and `male : divorced/separated_Mid-Career` (50% false negative rate), indicating systematic discrimination against specific demographic combinations.

Mitigation Strategies

1. Class Imbalance Mitigation

- **SMOTE (Synthetic Minority Over-sampling Technique):** Generate synthetic samples for the minority class (`credit_risk=0`) to achieve balanced 50-50 distribution, improving model sensitivity to good credit risk cases
- **Class Weighting:** Implement inverse class frequency weighting in Random Forest (`class_weight="balanced"`) to penalize misclassification of minority class more heavily
- **Ensemble Methods:** Use `BalancedRandomForest` or `EasyEnsemble` to create multiple balanced subsets through under-sampling of majority class

2. Missing Data Mitigation

- **Column Elimination:** Drop other_debtors entirely due to 90.7% missing rate exceeding typical usability thresholds (80-90%)
- **Strategic Imputation:** For other_installment_plans, create binary indicator (missing vs. non-missing) since "none" may represent meaningful absence of additional plans rather than true missing data
- **Placeholder Investigation:** Investigate amount=999 occurrence (0.1%) to determine if it represents missing/invalid data requiring imputation or exclusion

3. Fairness Mitigation

- **Reweighting:** Apply instance-level reweighting to balance positive prediction rates across demographic groups, ensuring statistical parity difference below 0.1
- **Adversarial Debiasing:** Implement adversarial networks to remove demographic information from learned representations while preserving credit risk predictive power
- **Threshold Optimization:** Adjust classification thresholds per demographic group to equalize false negative rates, ensuring qualified candidates from underrepresented groups receive equitable consideration
- **Intersectional Fairness Constraints:** Implement multi-group fairness constraints that optimize for worst-performing intersectional groups rather than aggregate metrics

Priority Order

1. **Class Imbalance Mitigation** - Highest priority due to direct impact on model performance and generalization capability. Addressing the 70-30 split will improve overall accuracy and minority class detection.
2. **Missing Data Column Elimination** - Second priority because other_debtors and other_installment_plans severely compromise data quality and may introduce systematic bias if retained without proper handling.
3. **Fairness Intersectional Constraints** - Third priority requiring careful implementation to avoid degrading overall performance while ensuring equitable outcomes across demographic groups.
4. **Threshold Optimization and Reweighting** - Fourth priority for fine-tuning after core model improvements are implemented, focusing on reducing disparate impact without sacrificing predictive accuracy.

Expected Impact

Performance Improvements

- **Balanced Accuracy:** Expected increase from current 76% to 82-85% through SMOTE and class weighting, with particular improvement in minority class recall (currently 42.67%)
- **F1-Score Enhancement:** Macro F1 should improve from 0.678 to 0.75-0.80 with better balance between precision and recall across both classes

- **Minority Class Detection:** Recall for credit_risk=0 should increase from 42.67% to 70-75%, significantly reducing false negatives for good credit risk applicants

Fairness Improvements

- **Statistical Parity Difference:** Reduction from 0.413 to below 0.1 for personal_status_sex, and from 0.1939 to below 0.05 for age through reweighting and threshold optimization
- **Disparate Impact:** Improvement from 0.5476 to above 0.8 threshold for single attributes, and from 0.6667 to above 0.8 for intersectional groups
- **False Negative Rate Equalization:** Reduction of FNR disparity from 3.75x ratio to below 1.5x across all demographic groups, particularly improving male : divorced/separated_Mid-Career from 50% to below 20%

Risk Mitigation

- **Model Stability:** Eliminating extreme missing data columns reduces variance and overfitting risks, improving generalization to new data
- **Regulatory Compliance:** Fairness improvements ensure compliance with anti-discrimination regulations and reduce legal exposure
- **Business Impact:** More equitable credit decisions expand eligible applicant pool while maintaining risk control through improved overall accuracy

Stage 6: Bias Mitigation

Status: success Applied Methods: Reweighting, SMOTE

Reweighting

Mitigation Results

- **Technique:** Reweighting (Balanced + Fair)
- **Dataset Size:** 1,000 → 1,000 (+0.0%)

Evaluation ML Model (Reweighting)

- **Algorithm:** Random Forest
- **Test Size:** 0.25
- **Accuracy:** 0.7320
- **Parameters:** `n\_estimators=100`, `max\_depth=None`

Evaluated Fairness Metrics

Sensitive Attribute	Stat Parity Diff	Disparate Impact	Highest Rate Group	Lowest Rate Group
personal + status + sex	0.2130	0.7667	male : married/ widowed	male : divorced/ separated
age	0.1415	0.8455	Early-Middle	Young

personal + status + sex + age	0.3333	0.6667	female : divorced/ separated/ married_ Early-Middle	male : divorced/ separated_ Early-Middle
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Mitigation Scorecard

Metric	Before Mitigation	After Mitigation	Improved?	Diff
Imbalance Ratio	2.33	1.00	Yes	-1.33
personal_status_sex (Stat Parity)	0.4130	0.2130	Yes	+0.2000
personal_status_sex (Disp Impact)	0.5476	0.7667	Yes	+0.2191
age (Stat Parity)	0.1939	0.1415	Yes	+0.0524
age (Disp Impact)	0.7854	0.8455	Yes	+0.0601
personal_status_sex_age_combined (Stat Parity)	1.0000	0.3333	Yes	+0.6667
personal_status_sex_age_combined (Disp Impact)	0.6667	0.6667	No	+0.0000

Agent Analysis

Analysis of Original vs. Mitigated Datasets

1. Was the bias mitigation effective?

Yes, but **only for the target variable**, not for sensitive attributes.

- The mitigation used **sample weighting** (not resampling or removal), so raw counts for sensitive attributes did not change.
- The **imbalance ratio** for the target variable improved from **2.33** ($\approx 70:30$) to **1.0** ($50:50$) in weighted terms, indicating effective rebalancing of class representation for modeling purposes.
- However, sensitive attributes (`personal\_status\_sex`, `age`) show **zero change** in counts or percentages, meaning group representation was not altered—only the influence of samples during training was adjusted.

2. What improved? (Specific metrics and percentages)

- **Target class balance (weighted):**
 - Class `1` (majority):
 - Original: 700 (70.0%)
 - Mitigated weighted: 499.5 (49.97%)
 - **Change:** -200.5 weighted count (-20.03 percentage points)
 - Class `0` (minority):
 - Original: 300 (30.0%)
 - Mitigated weighted: 500.0 (50.03%)
 - **Change:** +200.0 weighted count (+20.03 percentage points)

- **Imbalance ratio:**

- Original: 2.33

- Mitigated (weighted): 1.0

- **Improvement:** Yes (perfect balance in weighted space).

- **Key implication:**

During model training with these weights, the algorithm will treat the classes as balanced, reducing bias toward the majority class.

3. *What remained problematic?*

- **No change in sensitive attribute distributions:**

- ``personal_status_sex`` and ``age`` counts and percentages are identical between original and mitigated datasets.

- This means:

- No rebalancing across gender/age groups was performed.

- If the original data had unfair correlations between sensitive attributes and the target, those correlations remain in the data and may still affect model predictions unless explicitly addressed.

- **Reliance on weights without verification:**

- The note states that improvement will be realized only “during model training when weights are applied.”

- If the model does not properly use sample weights (e.g., some algorithms ignore them), the intended balance will not materialize.

- **Potential hidden bias:**

- Balancing the target overall does not guarantee fairness across subgroups (e.g., young females vs. older males). Without subgroup analysis, disparities may persist.

4. *Recommendations for further improvements*

1. **Apply sensitive-attribute-aware mitigation:**

- Use techniques such as **reweighting by intersection of sensitive attributes and target**, **resampling (SMOTE/undersampling) within subgroups**, or **adversarial debiasing** to reduce dependence between sensitive attributes and predictions.

2. **Verify weight application in modeling:**

- Confirm that the chosen model correctly uses ``sample_weight`` (e.g., in scikit-learn, most classifiers support it, but some preprocessing steps may ignore it).

- Monitor training with and without weights to ensure intended effect.

3. **Evaluate fairness metrics post-mitigation:**

- After training, measure:

- **Demographic parity difference**

- **Equalized odds / equal opportunity difference**

- **Disparate impact ratio** across sensitive groups (e.g., ``personal_status_sex`` categories).

- Ensure performance parity (e.g., similar TPR/FPR) across groups.

4. **Consider preprocessing with constraints:**

- Use methods like **Fairlearn’s reductions** or **AIF360’s reweighing** that explicitly optimize for fairness metrics during preprocessing or training.

5. **Document and test robustness:**

- Track how weights affect different slices of data.

- Perform ablation studies to isolate the effect of weighting vs. sensitive-attribute adjustments.

Bottom line::

The mitigation successfully balanced the target variable in weighted space, which should improve overall class fairness during training. However, without addressing sensitive attributes directly, subgroup fairness is not guaranteed. Next steps should focus on intersectional reweighting or constrained optimization to close remaining gaps.

Smote

Mitigation Results

- **Technique:** SMOTE
- **Dataset Size:** 1,000 → 1,400 (+40.0%)
- **Samples Added:** +400

Evaluation ML Model (SMOTE)

- **Algorithm:** Random Forest
- **Test Size:** 0.25
- **Accuracy:** 0.8543
- **Parameters:** `n\_estimators=100`, `max\_depth=None`

Evaluated Fairness Metrics

Sensitive Attribute	Stat Parity Diff	Disparate Impact	Highest Rate Group	Lowest Rate Group
personal + status + sex	0.4552	0.2022	male : single	male : divorced/ separated
age	0.1187	0.7577	Early-Middle	Mid-Career
personal + status + sex + age	0.6875	0.1616	male : single_ Early-Middle	male : divorced/ separated_ Early-Middle

Mitigation Scorecard

Metric	Before Mitigation	After Mitigation	Improved?	Diff
Imbalance Ratio	2.33	1.00	Yes	-1.33
personal_status_sex (Stat Parity)	0.4130	0.4552	No	-0.0422
personal_status_sex (Disp Impact)	0.5476	0.2022	No	-0.3454
age (Stat Parity)	0.1939	0.1187	Yes	+0.0752
age (Disp Impact)	0.7854	0.7577	No	-0.0277
personal_status_sex_age_ combined (Stat Parity)	1.0000	0.6875	Yes	+0.3125

personal_status_sex_age_combined (Disp Impact)	0.6667	0.1616	No	-0.5051
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Agent Analysis

Analysis of Bias Mitigation Results

1. Was the bias mitigation effective?

Yes — the mitigation successfully rebalanced the target variable and eliminated class imbalance, though it did so by up-sampling rather than re-weighting.

Why::

- The **imbalance ratio** improved from **2.33** to **1.0** (perfect balance).
- The target distribution shifted from **70% / 30%** to **50% / 50%**, reducing majority-class dominance.
- Sensitive-attribute groups were expanded in a way that increased representation without distorting the original counts of the majority class (class “1” remained at 700).

2. What improved? (specific metrics and percentages)

Metric	Original	Mitigated	Change
Dataset size	1,000	1,400	+400 (+40%)
Target class “0”	300 (30%)	700 (50%)	+400 (+20 pp)
Target class “1”	700 (70%)	700 (50%)	0 count, –20 pp
Imbalance ratio	2.33	1.0	Full balance achieved

Sensitive-attribute representation gains::

- **personal_status_sex:**
 - *male : divorced/separated*: 50 → 122 (+72, +144% relative)
 - *female : divorced/separated/married*: 310 → 443 (+133, +43% relative)
 - *male : single*: 548 → 656 (+108, +20% relative)
 - *male : married/widowed*: 92 → 179 (+87, +95% relative)
- **age:**
 - *Mid-Career*: 203 → 317 (+114, +56% relative)
 - *Late-Career*: 71 → 149 (+78, +110% relative)
 - *Early-Middle*: 315 → 424 (+109, +35% relative)
 - *Young*: 411 → 510 (+99, +24% relative)

All groups saw increased absolute counts, reducing under-representation.

3. What remained problematic?

- **No change in the original majority-class count:** Class “1” remained fixed at 700. While this prevents overfitting to synthetic/duplicate majority samples, it means the model's exposure to class “1” patterns is unchanged — if bias was encoded in those patterns, it may persist.
- **Up-sampling without weights:** Since ``uses_weights = false``, the mitigation likely added duplicated or synthetic rows. This can:
 - Inflate dataset size (+40%), increasing compute/time.
 - Risk overfitting to repeated patterns, especially for smaller sensitive groups (e.g., `*male : divorced/separated*` grew 144% but from a small base).
- **No fairness metrics reported:** We don't see changes in disparity measures (e.g., demographic parity difference, equalized odds) or model performance (accuracy, F1) by group. Without these, we can't confirm whether the mitigation improved fairness in predictions or just data balance.

4. Recommendations for further improvements

1. Add fairness and performance metrics by group

- Report demographic parity, equal opportunity, and F1/accuracy for each sensitive group before/after mitigation.
- This will clarify whether the data rebalancing translated to fairer predictions.

2. Consider hybrid rebalancing

- Use **sample weights** (``uses_weights = true``) instead of or alongside up-sampling to avoid inflating dataset size and reduce overfitting risk.
- Combine with **SMOTE or ADASYN** for minority class “0” to generate synthetic (rather than duplicated) samples.

3. Apply targeted augmentation for underrepresented intersections

- Groups like `*male : divorced/separated*` and `*Late-Career*` still have low absolute counts post-mitigation (122 and 149).
- Use targeted oversampling or data augmentation for these intersections to ensure robust representation.

4. Validate model calibration and bias in predictions

- Check if predicted probabilities are calibrated across groups.
- Run bias audits (e.g., disparate impact ratio, false positive/negative rate parity) to ensure no new biases were introduced.

5. Document the mitigation method explicitly

- Clarify whether up-sampling was random, SMOTE-based, or used other techniques.
- This affects reproducibility and helps diagnose overfitting.

Bottom line: The mitigation effectively balanced classes and improved representation, but further fairness-aware validation and potentially weight-based or synthetic sampling would strengthen robustness without inflating data size.

Report generated by Dataset Fairness Evaluation System